ECON 508
Problem set 5, question 2

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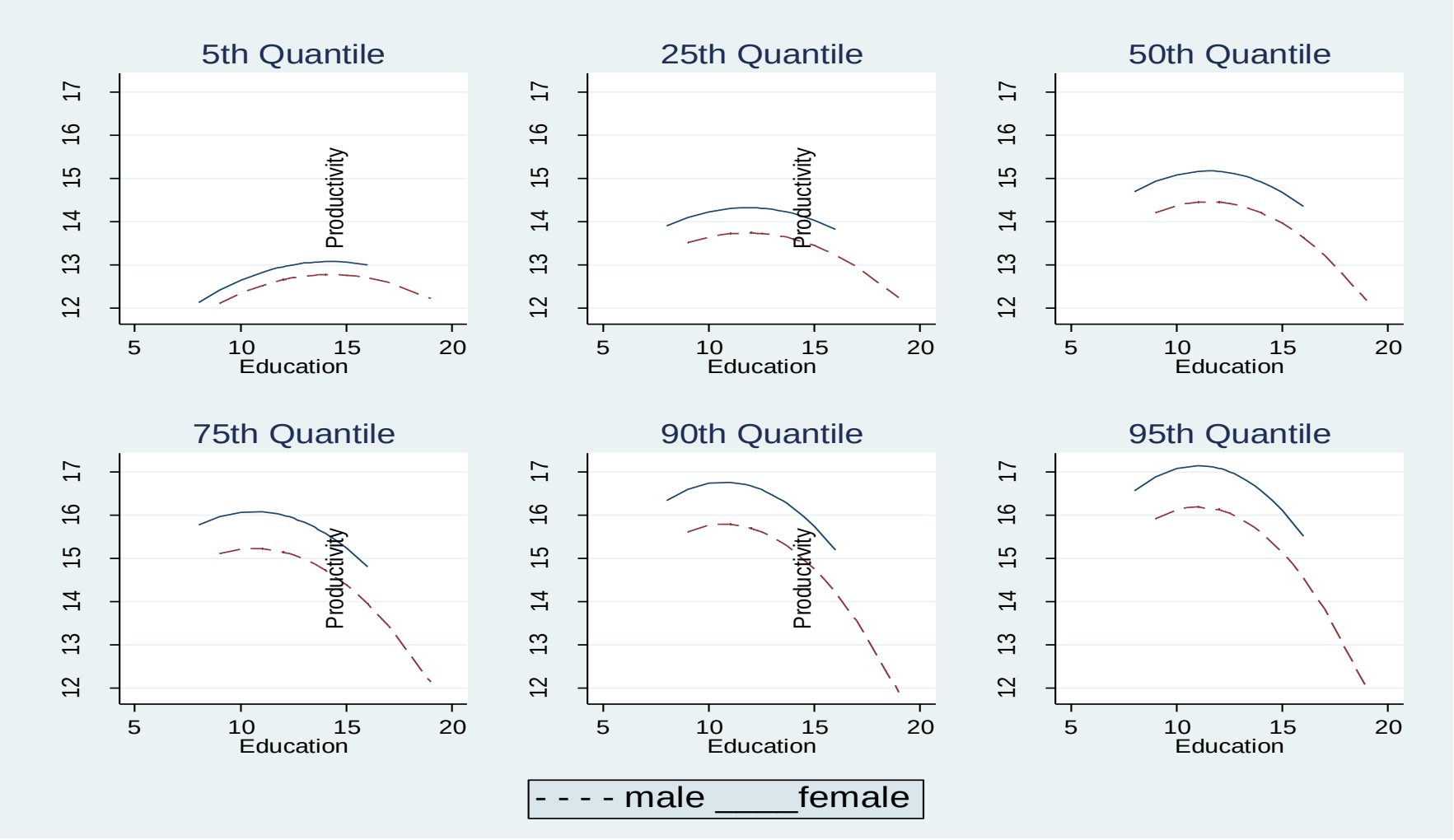
Quantile regression

$$Q_{y_i}(\tau|x_i) = \beta_0(\tau) + \beta_1(\tau)sex_i + \beta_2(\tau)dex_i + \beta_3(\tau)lex_i + \beta_4(\tau)lex_i^2 + u_i$$

	sex	dex	lex	lex^2	constant
Q5	-0.477*	0.108***	0.695	-0.0244	3.374
	(0.187)	(0.014)	(0.528)	(0.0196)	(3.615)
Q25	-0.758***	0.106***	0.690*	-0.0292**	5.557**
	(0.0959)	(0.0068)	(0.279)	(0.0109)	(1.818)
Q50	-0.897***	0.112***	0.928**	-0.0404**	4.924*
	(0.105)	(0.0072)	(0.347)	(0.0136)	(2.242)
Q75	-1.042***	0.115***	0.943*	-0.0443**	5.996*
	(0.126)	(0.0082)	(0.408)	(0.0158)	(2.676)
Q95	-1.175***	0.128***	1.447	-0.0657*	3.581
	(0.197)	(0.0125)	(0.821)	(0.0313)	(5.358)

Productivity estimated and years of education by Quantile



Hiring strategy I

firm chooses a cutoff of 14 for productivity.

Quantile\lex	8	9	10	11	12	13	14	15	16
5th	12.12776	12.40758	12.63856	12.8207	12.954	13.03846	13.07407	13.06085	12.99878
25th	13.89796	**14.09197**	**14.2276**	**14.30485**	**14.32372**	**14.28421**	**14.18632**	**14.03005**	13.8154
50th	**14.68877**	14.92945	15.08927	15.16822	15.16629	15.0835	14.91984	14.67531	**14.34992**
75th	15.78017	15.96967	16.07054	16.08278	16.00638	15.84136	15.5877	15.24542	14.8145
95th	16.56638	16.89613	17.09445	17.16134	17.09679	16.90081	16.57339	16.11454	15.52426

Quantile\lex	9	10	11	12	13	14	15	16	17
5th	12.10632	12.3373	12.51944	12.65273	12.73719	12.77281	12.75958	12.69752	12.58661
25th	13.50698	13.6426	13.71985	13.73872	13.69921	13.60132	13.44505	13.2304	12.95737
50th	**14.21512**	**14.37493**	**14.45388**	**14.45196**	**14.36917**	**14.20551**	13.96098	13.63558	13.22932
75th	15.11548	15.21635	15.22859	15.1522	14.98717	14.73352	**14.39123**	13.96031	13.44076
95th	15.92838	16.12671	16.19359	16.12904	15.93306	15.60565	15.1468	**14.55652**	13.8348

Hiring strategy II

x is 1 if $y \geq 13$ and otherwise 0

$$logit\left(P(x=1)\right) = \gamma_0 + \gamma_1 sex + \gamma_2 dex + \gamma_3 lex + \gamma_4 lex^2$$

	x
sex	-0.800**
	(-2.95)
dex	0.176***
	(7.72)
lex	0.705
	(0.8)
lex2	-0.0354
	(-1.04)
_cons	-8.219
	(-1.44)

t statistics in parentheses
* p<0.05, **p<0.01, *** p<0.001